

10. PowerShell Commands for Active Directory and Connecting to Microsoft Entra ID

Aim

To perform basic user, computer, and group administration in Active Directory and retrieve user/group information from Microsoft Entra ID using Microsoft Graph PowerShell.

Active Directory Module

```
Import-Module ActiveDirectory
```

AD Users

```
Get-ADUser -Filter *  
Get-ADUser -Identity "john"  
Get-ADUser -Identity "john" -Properties *
```

AD Computers and Groups

```
Get-ADComputer -Filter *  
Get-ADGroup -Filter *  
Get-ADGroupMember "Domain Admins"
```

AD Account Searches

```
Search-ADAccount -AccountDisabled  
Search-ADAccount -LockedOut
```

Microsoft Entra ID / Azure AD

Microsoft Entra ID is the current name for Azure Active Directory. Microsoft Graph PowerShell is the recommended approach for many Entra ID administration tasks.

```
Install-Module Microsoft.Graph -Scope CurrentUser  
Connect-MgGraph -Scopes "User.Read.All"  
Get-MgUser -All  
Get-MgUser -All | Select-Object DisplayName, UserPrincipalName, AccountEnabled  
Get-MgUser -UserId "user@domain.com"  
Get-MgGroup -All
```

Use only accounts, domains, and resources for which you have authorization. Directory commands may require administrator consent or appropriate delegated/application permissions.

Final Integrated Practical Exercise

Create a PowerShell script named `system_report.ps1` that performs the following tasks:

- Display the computer name and current username.
- Display operating system information using `Get-CimInstance`.
- Display IP configuration.
- Count the number of running processes.
- Display the top five processes by memory usage.
- Display the number of running services.

- Export selected process information to a CSV file.

```
$report = [PSCustomObject]@{  
    ComputerName = $env:COMPUTERNAME  
    UserName = $env:USERNAME  
    RunningProcesses = (Get-Process | Measure-Object).Count  
    RunningServices = (Get-Service | Where-Object {$_.Status -eq "Running"} | Measure-Object).Count  
}  
  
$report | Format-List  
  
Get-Process |  
    Sort-Object WorkingSet64 -Descending |  
    Select-Object -First 5 Name, Id, WorkingSet64 |  
    Export-Csv ".\TopProcesses.csv" -NoTypeInformation
```

Safety and Good Practice

- Run administrative commands only on systems you are authorized to manage.
- Test scripts in a lab environment before using them on production systems.
- Review commands that modify, delete, start, stop, or configure resources before execution.
- Do not store passwords, access tokens, or other credentials directly in scripts.
- Use Get-Help and command documentation to understand unfamiliar commands.